GEO 3452 | Introduction to Medical Geography

Class Number: 13445

Summer 2026
Delivery Method: Online

Instructor Information

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Office Hours
Mondays 9am - 11am - zoom link available on course Canvas page
Preferred Method of Communication: Email

Office Location

Zoom link available on course Canvas page

Course Details

Catalog Description: Medical geography deals with human-environment interactions and the influence of these interactions on public health. Provides a broad and comprehensive survey of geographic approaches in medical studies.
Pre- and Co-Requisites: Prereq: sophomore standing or higher; entry-level knowledge of statistics (STA 2023 or GEO 3162C or equivalent) recommended.
Credit Hours: 3
Course Fees: $10.50

Additional Course Description

Medical geography deals with human-environment interactions and the influence of these interactions on public health. Provides a broad and comprehensive survey of geographic approaches in medical studies.

Recommended Materials

HEALTH AND MEDICAL GEOGRAPHY
ISBN: 1462520065
Authors: EMCH, ROOT, CARREL
Publisher: THE GUILFORD PRESS
Edition: 4TH

Health and Medical Geography 4E (Kortext eBook)
Publisher: Guilford Publications
All Access: This material is available on All Access

A Note on Materials

The readings for this course will be made available via Canvas. The textbook ``HEALTH AND MEDICAL GEOGRAPHY`` is a recommended reading but is not required. Note From Bookstore: All Access will be available until 6/19/2026. You may opt-in on https://www.bsd.ufl.edu/allaccess ~~

Course Goals and Objectives

This course introduces fundamental concepts and approaches of medical geography to understand how interactions between humans and their environments shape patterns of health and disease.

In this course students will:1) apply basic concepts, principles, and methods that are widely used in medical geography, 2) investigate health problems with a systematic approach and spatial analysis methods, and 3) analyze real-world health problems using geographic information systems software.

By the end of the course, students will be able to analyze the spatial distribution of health outcomes and apply geographic perspectives to contemporary medical and health-related issues.

Expectations and Student Learning Outcomes

A student who successfully completes this course will be able to:

• Demonstrate competence in the terminology, concepts, theories and methodologies used within the discipline.

• Logically analyze information from multiple perspectives and develop reasoned solutions to problems within the discipline.

• Students communicate knowledge, ideas and reasoning clearly and effectively in written and oral forms appropriate to the discipline.

Methods of Evaluation

Discussions - 20%

Quizzes - 20%

GIS Labs - 20%

Midterm - 10%

Final Presentation - 15%

Final Paper - 15%

Grading Scale

Letter grade and percentage

Letter	Percentage Value
A	94 - 100%
A -	90 - 93%
B +	87 - 89%
B	83 - 86%
B -	80 - 82%
C +	77 - 79%
C	73 - 76%
C -	70 - 72%
D +	67 - 69%
D	63 - 66%
D -	60 - 62%
E	59% and below

Course Schedule

Week 1: Course overview, introduction to medical geography, basic concepts and principles

Week 1 Readings:

Hippocrates. (1886). The genuine works of Hippocrates (F. Adams, Trans.). New York, NY: Wood.

Week 2: Measuring diseases, methodology design, inferring causal relationships

Week 2 Readings:

Ressing, M., Blettner, M., & Klug, S. J. (2010). Data analysis of epidemiological studies: Part 11 of a series on evaluation of scientific publications. Deutsches Arzteblatt International, 107(11), 187-192.

Week 3: Maps and GIS, vector-borne disease, zoonoses

Week 3 Readings:

Richardson, D. B., Volkow, N. D., Kwan, M. P., Kaplan, R. M., Goodchild, M. F., & Croyle, R. T. (2013). Spatial turn in health research. Science, 339(6126), 1390-1392.

Townes, L. R., Mwandama, D., Mathanga, D. P., & Wilson, M. L. (2013). Elevated dry-season malaria prevalence associated with fine-scale spatial patterns of environmental risk: A case–control study of children in rural Malawi. Malaria Journal, 12(407). https://doi.org/10.1186/1475-2875-12-407.

Chapter 2 of Emch, M., Root, E. D., & Carrel, M. (2017). Health and medical geography (4th ed.). New York, NY: The Guilford Press

Week 4: Climate change and global health, air pollution and human health

Week 4 Readings:

Maantay, J. (2007). Asthma and air pollution in the Bronx: Methodological and data considerations in using GIS for environmental justice and health research. Health & Place, 13(1), 32-56.https://doi.org/10.1016/j.healthplace.2005.09.009

Young, A. (2012, April 25). Long-gone lead factories leave poisons in nearby yards. USA Today. Retrieved from https://www.usatoday.com/

Week 5: Spatial diffusion of diseases, human behavior and health, spatial disparities in health care

Week 5 Readings:

Berndt, D. J., Fisher, J. W., Rajendrababu, R. V., & Studnicki, J. (2003, January). Measuring healthcare inequities using the Gini index. Proceedings of the 36th Annual Hawaii International Conference on System Sciences, Big Island, HI, 1-10. https://doi.org/10.1109/HICSS.2003.1174353.

Week 6: Wrap-up

Week 6 Readings: No readings

Schedule of Assignments

Due Dates	Assignments	Type of Assignment	Points
5/17/26	Orientation quiz	Quiz	100
5/17/26	Week 1 quiz	Quiz	100
5/17/26	Discussion: Introduce Yourself	Discussion post	120
5/17/26	Lab Setup Quiz	Quiz	100
5/17/26	GIS Lab 1	GIS lab	150
5/24/26	Week 2 Quiz	Quiz	100

Due Dates	Assignments	Type of Assignment	Points
5/24/26	Discussion: Final Project Topic Selection & Practice	Discussion post	120
5/24/26	GIS Lab 2	GIS lab	150
5/31/26	Week 3 Quiz	Quiz	100
5/31/26	Discussion: Midterm Study Guide	Discussion post	120
5/31/26	GIS Lab 3	GIS lab	150
6/3/26	Midterm Exam	Exam	300
6/7/26	GIS Lab 4	GIS lab	150
6/7/26	Discussion: Case Study #4 Environmental Justice	Discussion	120
6/14/26	Week 5 Quiz	Quiz	100
6/14/26	Discussion: Case Study 7: Spatial Disparities in Health Care	Discussion post	120
6/14/26	Final Presentation	Presentation	450

Due Dates	Assignments	Type of Assignment	Points
6/17/26	Final Term Paper	Paper	450

University Policies and Resources

Information about grading policies, support for students with disabilities, course evaluations, the Honor Code, and other course policies and campus resources can be found on the Syllabus Policies page.

Attendance Policy

Excused and Unexcused Absences

Students may only participate in classes if they are registered officially or approved to audit with evidence of having paid audit fees. The Office of the University Registrar provides official class rolls to instructors.

Students are responsible for satisfying all academic objectives as defined by the instructor. Absences count from the first-class meeting.

Acceptable reasons for absence from or failure to engage in class include illness; Title IX-related situations; serious accidents or emergencies affecting the student, their roommates, or their family; special curricular requirements (e.g., judging trips, field trips, professional conferences); military obligation; severe weather conditions that prevent class participation; religious holidays; participation in official university activities (e.g., music performances, athletic competition, debate); and court-imposed legal obligations (e.g., jury duty or subpoena). Other reasons (e.g., a job interview or club activity) may be deemed acceptable if approved by the instructor.

For all planned absences, a student in a situation that allows an excused absence from a class, or any required class activity must inform the instructor as early as possible prior to the class. For all unplanned absences because of accidents or emergency situations, students should contact their instructor as soon as conditions permit.

Students shall be permitted a reasonable amount of time to make up the material or activities covered during absence from class or inability to engage in class

activities because of the reasons outlined above.

If a student does not participate in at least one of the first two class meetings of a course or laboratory in which they are registered, and they have not contacted the department to indicate their intent, the student can be dropped from the course. Students must not assume that they will be dropped, however. The department will notify students if they have been dropped from a course or laboratory.

The university recognizes the right of the instructor to make attendance mandatory and require documentation for absences (except for religious holidays), missed work, or inability to fully engage in class. After due warning, an instructor can prohibit further attendance and subsequently assign a failing grade for excessive absences.

Religious Holidays Guidelines

At the University of Florida, students and faculty work together to allow students the opportunity to observe the holy days of their faith. A student should inform the faculty member of the religious observances of their faith that will conflict with class attendance, with tests or examinations, or with other class activities prior to the class or occurrence of that test or activity. The faculty member is then obligated to accommodate that particular student's religious observances. Because students represent a myriad of cultures and many faiths, the University of Florida is not able to assure that scheduled academic activities do not conflict with the holy days of all religious groups. Accordingly, individual students should make their need for an excused absence known in advance of the scheduled activities.

The Florida Board of Education and state law govern university policy regarding observance of religious holidays.

Guidelines

- Students, upon prior notification to their instructors, shall be excused from class or other scheduled academic activity to observe a religious holy day of their faith.
- Students shall be permitted a reasonable amount of time to make up the material or activities covered in their absence.
- Students shall not be penalized due to absence from class or other scheduled academic activity because of religious observances.

If a faculty member is informed of or is aware that a significant number of students are likely to be absent from class because of a religious observance, the faculty member should not schedule a major exam or other academic event at that time.

A student who is to be excused from class for a religious observance is not required to provide a second party certification of the reason for the absence. Furthermore, a student who believes that they have been unreasonably denied an education benefit due to religious beliefs or practices may seek redress through the student grievance procedure.

Absence due to Illness

A student who is absent from class or any required class-related activity because of illness should contact their instructor, if feasible, as early as possible prior to the missed class or activity.

Students shall be permitted a reasonable amount of time to make up the material or activities covered during an excused absence.

Students should contact their college by the deadline to drop a course for medical reasons. Students can petition the Dean of Students Office to drop a course for medical reasons. The university's policy regarding medical excuse from classes is maintained by the Student Health Care Center.

Twelve-Day Rule

Students who participate in university-sponsored athletic or scholarly activities are permitted to be absent 12 scholastic days per semester without penalty. A scholastic day is any day on which regular class work is scheduled as defined in the approved university calendar. More Info

The student or student's advisor must notify the instructor as early as possible prior to the anticipated absence to allow ample time for accommodations. Instructors must be flexible and not penalize students when re-scheduling during-term and final exams, class assignments, and other required activities and must follow the UF Attendance Policy herein and UF Examination Policies. As noted in the UF Examination Policies, during-term exams should be re-scheduled no later than before the end of the semester, while final exams no later than 90 days after the originally scheduled exam time. However, instructors are encouraged to re-schedule final and during-term exams, assignments, and other activities as soon

as possible after the last day of the absence and must not penalize the student in any way. More Info

A group's schedule that requires absence of more than 12 scholastic days should be adjusted so that no student is absent from campus more than 12 scholastic days. Students who previously have been warned in writing by their instructor about the impact of absences on their individual class performance should not incur additional absences, even if they have not been absent 12 scholastic days. The student is responsible to maintain satisfactory academic performance and attendance.